

Stereo Distance Measurement with a Per-Camera Γ -Grid Model

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Abstract

We measure metric distance from two calibrated cameras with a deliberately small, fully transparent model: each camera is described by a $5 \times 5 \times 2$ grid Γ of *effective inverse focal lengths* interpolated bilinearly over the image, and the rig by a baseline $\mathbf{d}$ and a relative rotation $R(\omega)$. All gradients are derived by hand and the model is fitted with Adam on a robust relative-disparity loss; no automatic differentiation, SciPy or neural networks are used. On 27 hand-labelled real points the leave-one-out error halves with respect to the first version of the system (10.1 % $\rightarrow$ 5.1 % MAPE); on a simulated rig with Brown–Conrady distortion, where ground truth is exact, the Γ grid reaches 0.5–1 % error, four to five times better than a pinhole model with the same extrinsics.

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1 Pixel and normalised coordinates

A pixel is (u, v) on a $W \times H$ sensor (2592×1944). Optimising in pixels is badly scaled, so we map to normalised coordinates centred on $(C_X, C_Y) = (W/2, H/2)$:

$$\tilde{u} = \frac{u - C_X}{C_X}, \quad \tilde{v} = \frac{v - C_Y}{C_Y} \quad \in [-1, 1]. \quad (1)$$

2 The ray equation

Every pixel defines a ray from the optical centre. In our model

$$\mathbf{r}(u, v) = \begin{bmatrix} \tilde{u} \gamma_x(u, v) \\ \tilde{v} \gamma_y(u, v) \\ 1 \end{bmatrix}. \quad (2)$$

For an ideal pinhole with focal length f (pixels), $\mathbf{r} = [(u - C_X)/f, (v - C_Y)/f, 1]^\top$, i.e. $\gamma_x = C_X/f$ and $\gamma_y = C_Y/f$ everywhere. Letting γ vary over the image absorbs lens distortion without a polynomial distortion model.

Why two channels. Version 1 used a single γ for both axes. Since $C_X \neq C_Y$ ($C_X/C_Y = 4/3$), a single value cannot equal both C_X/f and C_Y/f : the vertical ray component was scaled by $4/3$. The two-channel grid $\Gamma \in \mathbb{R}^{5 \times 5 \times 2}$ removes this error; the nominal value is $\Gamma_0 = [C_X/f_0, C_Y/f_0]$ with $f_0 = 2880$ px.

3 Bilinear interpolation

Grid nodes sit at $u = j c_w$, $v = i c_h$ with $c_w = W/4$, $c_h = H/4$. With $f_x = u/c_w$, $f_y = v/c_h$, cell $(i_y, i_x) = (\lfloor f_y \rfloor, \lfloor f_x \rfloor)$ (clamped to the grid) and $t_x = f_x - i_x$, $t_y = f_y - i_y$:

$$\gamma(u, v) = \sum_{k=1}^4 w_k \Gamma_{n_k}, \quad (w_1, \dots, w_4) = ((1 - t_x)(1 - t_y), t_x(1 - t_y), (1 - t_x)t_y, t_x t_y). \quad (3)$$

The weights sum to one, reproduce any field linear in (u, v) exactly, and make γ C^0 -continuous; evaluation is $O(1)$ (four multiply-adds). Crucially, $\partial\gamma/\partial\Gamma_{n_k} = w_k$, which makes all gradients cheap.

4 Stereo geometry

4.1 Rig model

Let P_L, P_R be the coordinates of one 3-D point in the left and right camera frames. The rig is described by

$$P_L = R(\boldsymbol{\omega}) P_R + \mathbf{d}, \quad \mathbf{d} = [d_x, d_y, 0]^\top, \quad (4)$$

with $R = R_z(\omega_{\text{roll}}) R_y(\omega_{\text{yaw}}) R_x(\omega_{\text{pitch}})$. Version 1 assumed $R = I$. The real data contradict this: the horizontal disparity follows $\Delta u \approx 575/Z + 61$ px (a constant offset $\Rightarrow$ yaw), and the vertical disparity grows linearly with u (+44 px at the right border, -32 px at the left $\Rightarrow$ roll). A multiplicative γ can produce neither an additive offset at $\tilde{u} = 0$ nor an $x \leftrightarrow y$ coupling, so three angles are added.

For small angles $R \approx I + [\boldsymbol{\omega}]_\times$, and the exact partial derivatives are

$$\frac{\partial R}{\partial \omega_{\text{pitch}}} = R_z R_y R'_x, \quad \frac{\partial R}{\partial \omega_{\text{yaw}}} = R_z R'_y R_x, \quad \frac{\partial R}{\partial \omega_{\text{roll}}} = R'_z R_y R_x, \quad (5)$$

where R'_* is the element-wise derivative of the elementary rotation.

4.2 Rectification in ray space

The right ray is rotated into the left frame and perspective-normalised:

$$\mathbf{w} = R\mathbf{r}_R, \quad \hat{\mathbf{r}}_R = \mathbf{w}/w_z. \quad (6)$$

Writing $P_L = Z\mathbf{r}_L$ and $P_R \propto \mathbf{r}_R$, eq. (4) becomes

$$Z\mathbf{r}_L - Z'\hat{\mathbf{r}}_R = \mathbf{d}. \quad (7)$$

The z -row gives $Z = Z'$ (because $d_z = 0$), hence the *disparity relations*

$$r_{L,x} - \hat{r}_{R,x} = \frac{d_x}{Z}, \quad r_{L,y} - \hat{r}_{R,y} = \frac{d_y}{Z}. \quad (8)$$

4.3 Triangulation by ordinary least squares

Eq. (7) is three equations in two unknowns, $A\mathbf{x} = \mathbf{d}$ with $A = [\mathbf{r}_L \quad -\hat{\mathbf{r}}_R] \in \mathbb{R}^{3 \times 2}$, $\mathbf{x} = [Z, Z']^\top$. Rays rarely intersect, so we solve the normal equations

$$A^\top A\mathbf{x} = A^\top \mathbf{d} \implies Z = \frac{s_{22}b_1 - s_{12}b_2}{s_{11}s_{22} - s_{12}^2}, \quad (9)$$

with $s_{ij} = \mathbf{a}_i \cdot \mathbf{a}_j$ and $b_i = \mathbf{a}_i \cdot \mathbf{d}$ (closed-form 2×2 inverse, vectorised over all points).

4.4 The same problem by batch gradient descent

Minimising $f(\mathbf{x}) = \|A\mathbf{x} - \mathbf{d}\|^2$ with $\nabla f = 2(A^\top A\mathbf{x} - A^\top \mathbf{d})$ and step $\eta = 1/L$, $L = 2\lambda_{\max}(A^\top A)$, contracts the error by $(1 - 1/\kappa)$ per step, $\kappa = \lambda_{\max}/\lambda_{\min}$. For distant points the rays are nearly parallel: $\lambda_{\min} \approx \frac{1}{2}(d_x/Z)^2$, and κ reaches 10^3 – 10^4 . Nesterov momentum $\beta = (\sqrt{\kappa} - 1)/(\sqrt{\kappa} + 1)$ reduces the iteration count to $O(\sqrt{\kappa})$ (~ 600 – 2000 iterations in practice). The iterative answer converges to the OLS answer to 10^{-8} relative precision (tested), which is why OLS is the authoritative path and GD is kept as a check (`StereoModel.triangulate_gd`).

4.5 Why depth accuracy degrades with distance

From eq. (8), $Z = d_x/D$ with $D = r_{L,x} - \hat{r}_{R,x}$, so $\partial Z/\partial D = -Z^2/d_x$: a fixed pixel error causes a depth error that grows with Z^2 .

5 Loss function

5.1 Relative disparity and relative epipolar error

Instead of $(Z_{\text{pred}} - Z)^2$, which inherits the Z^2 blow-up above, we use

$$e_x = D_x \frac{Z}{d_x} - 1, \quad D_x = r_{L,x} - \hat{r}_{R,x}, \quad (10)$$

$$e_y = D_y \frac{Z}{d_x} - \frac{d_y}{d_x}, \quad D_y = r_{L,y} - \hat{r}_{R,y}. \quad (11)$$

e_x is the relative disparity error ($\approx$ relative depth error): 5% at 1 m costs the same as 5% at 5 m. e_y measures the vertical (epipolar) misalignment *in the same units*, which in v1 was an absolute ray quantity an order of magnitude smaller than e_x (two orders after squaring) and therefore nearly ignored.

5.2 Huber robustness

Hand-clicked labels contain outliers, so each residual is wrapped in the Huber function

$$\rho_\delta(e) = \begin{cases} \frac{1}{2}e^2, & |e| \leq \delta \\ \delta(|e| - \frac{1}{2}\delta), & |e| > \delta \end{cases}, \quad \psi(e) = \rho'_\delta(e) = \text{clip}(e, -\delta, \delta), \quad (12)$$

with $\delta = 0.05$ (5 %). Beyond δ a point's gradient is capped, so a mis-click cannot dominate the fit.

5.3 Full objective

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N [\rho_\delta(e_{x,i}) + \lambda_y \rho_\delta(e_{y,i})] + \frac{\lambda_a}{2} \sum_{c \in \{L,R\}} \|\Gamma_c - \Gamma_0\|^2 + \lambda_s \sum_{c \in \{L,R\}} E_{\text{mem}}(\Gamma_c), \quad (13)$$

with $\lambda_y = 0.5$, $\lambda_a = 3 \cdot 10^{-3}$ and $\lambda_s = 0.3$ (selected by leave-one-out cross-validation, §12).

6 Analytic gradients

Let $g_x = \psi(e_x)/N$, $g_y = \lambda_y \psi(e_y)/N$ and $s = Z/d_x$.

Baseline. From (10)–(11): $\partial e_x / \partial d_x = -(e_x + 1)/d_x$, $\partial e_y / \partial d_x = -e_y/d_x$, $\partial e_y / \partial d_y = -1/d_x$, so

$$\frac{\partial \mathcal{L}}{\partial d_x} = - \sum_i \frac{g_x(e_x + 1) + g_y e_y}{d_x}, \quad \frac{\partial \mathcal{L}}{\partial d_y} = - \sum_i \frac{g_y}{d_x}. \quad (14)$$

Left grid. $r_{L,x} = \tilde{u}\gamma_x$, $\partial e_x / \partial r_{L,x} = s$ and $\partial \gamma_x / \partial \Gamma_{n_k,x} = w_k$, hence

$$\frac{\partial \mathcal{L}}{\partial \Gamma_{L,n_k,x}} = g_x s \tilde{u}_L w_k, \quad \frac{\partial \mathcal{L}}{\partial \Gamma_{L,n_k,y}} = g_y s \tilde{v}_L w_k. \quad (15)$$

Through the perspective division. With $\mathbf{a} = \partial \mathcal{L} / \partial \hat{\mathbf{r}}_{R,xy} = (-g_x s, -g_y s)$ and $\hat{r}_x = w_x/w_z$, $\hat{r}_y = w_y/w_z$:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{w}} = \left[\frac{a_1}{w_z}, \frac{a_2}{w_z}, -\frac{a_1 w_x + a_2 w_y}{w_z^2} \right]^\top. \quad (16)$$

Right grid and rotation. Since $\mathbf{w} = R \mathbf{r}_R$,

$$\frac{\partial \mathcal{L}}{\partial \mathbf{r}_R} = R^\top \frac{\partial \mathcal{L}}{\partial \mathbf{w}}, \quad \frac{\partial \mathcal{L}}{\partial \omega_j} = \sum_i \left(\frac{\partial \mathcal{L}}{\partial \mathbf{w}_i} \right)^\top \frac{\partial R}{\partial \omega_j} \mathbf{r}_{R,i}, \quad (17)$$

and the right grid receives $(\partial \mathcal{L} / \partial r_{R,x}) \tilde{u}_R w_k$ and $(\partial \mathcal{L} / \partial r_{R,y}) \tilde{v}_R w_k$ exactly as the left one.

Regularisers. $\nabla (\frac{\lambda_a}{2} \|\Gamma - \Gamma_0\|^2) = \lambda_a (\Gamma - \Gamma_0)$ and $\nabla E_{\text{mem}} = \mathbf{L} \Gamma$ (§8).

Verification. Every component of the gradient is compared with central finite differences in the test-suite; the largest relative discrepancy is $1.2 \cdot 10^{-7}$.

7 Adam optimiser

For each parameter block $\theta \in \{\Gamma_L, \Gamma_R, \mathbf{d}, \boldsymbol{\omega}\}$ with gradient g_t :

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \quad v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2, \quad (18)$$

$$\hat{m}_t = m_t / (1 - \beta_1^t), \quad \hat{v}_t = v_t / (1 - \beta_2^t), \quad (19)$$

$$\theta_t = \theta_{t-1} - \eta_t \hat{m}_t / (\sqrt{\hat{v}_t} + \epsilon), \quad \eta_t = \eta [\alpha + (1 - \alpha) \frac{1}{2} (1 + \cos(\pi t / T))], \quad (20)$$

with $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, a cosine schedule to $\alpha = 5\%$, per-block rates $\eta_\Gamma = \eta_d = 2 \cdot 10^{-3}$, $\eta_\omega = 10^{-3}$ and $T = 600$ epochs. Adam's per-coordinate normalisation matters here because Γ (dimensionless), $\mathbf{d}$ (metres) and $\boldsymbol{\omega}$ (radians) have very different gradient scales. After each step Γ is clipped to $[0.05, 5]$ and $d_x \geq 0.02$ m.

8 Regularisation, gauge and initialisation

8.1 Anchor prior

$\lambda_a(\Gamma - \Gamma_0)$ pulls nodes without data back to the pinhole value, so grid cells without calibration points cannot drift to unphysical values.

8.2 Membrane smoothness

Real lenses distort smoothly. We penalise the membrane (Dirichlet) energy on the 4-connected grid, per channel,

$$E_{\text{mem}}(\Gamma) = \frac{1}{2} \sum_{(i,j) \in \mathcal{E}} (\Gamma_i - \Gamma_j)^2, \quad \nabla E_{\text{mem}} = \mathbf{L} \Gamma, \quad (21)$$

where $\mathbf{L}$ is the graph Laplacian: $4\Gamma_{ij} - \sum \text{neighbours in the interior}$, and the true node degree (2 or 3) on the border. (Version 1 applied the interior stencil only, which is not the gradient of any energy.)

8.3 Gauge fixing

Scaling every γ and d_x by the same factor leaves (8) unchanged (exactly without rotation, approximately with it), so the absolute scale is not identifiable. We fix it by freezing the centre node of Γ_L at Γ_0 . Consequently d_x is the baseline *expressed at the nominal focal length* f_0 ; it equals the physical baseline only if the true focal length is f_0 . The centre of Γ_R stays free so the cameras may differ.

8.4 Closed-form warm start

Linearising $R \approx I + [\boldsymbol{\omega}]_\times$ with nominal Γ gives two equations per point that are *linear* in $(d_x, d_y, \omega_p, \omega_y, \omega_r)$:

$$r_{L,x} - r_{R,x} = d_x / Z + \omega_y - \omega_r r_{R,y}, \quad (22)$$

$$r_{L,y} - r_{R,y} = d_y / Z - \omega_p + \omega_r r_{R,x}, \quad (23)$$

solved by weighted least squares (rows scaled by Z so residuals are relative). This DLT-style initialisation alone gives 5.2% in-sample error; Adam then refines all 105 parameters (2×50 grid values, d_x , d_y and three angles; 103 free after gauge fixing).

9 Epipolar geometry and correspondence search

9.1 Forward projection and grid inversion

For a left pixel and a candidate depth Z : $P = Z\mathbf{r}_L$, $P_R = R^\top(P - \mathbf{d})$, $(t_x, t_y) = (P_{R,x}, P_{R,y})/P_{R,z}$. The right pixel solves $\tilde{u}\gamma_x(u, v) = t_x$, $\tilde{v}\gamma_y(u, v) = t_y$, found by the fixed-point iteration

$$u \leftarrow C_X + C_X t_x / \gamma_x(u, v), \quad v \leftarrow C_Y + C_Y t_y / \gamma_y(u, v), \quad (24)$$

which converges in a few steps because Γ varies slowly. Sweeping Z uniformly in $1/Z$ traces the (curved, tilted) epipolar curve. A version-1 style search along the image *row* misses the true match by up to ~ 40 px on this rig because of the roll.

9.2 Zero-mean normalised cross-correlation

For a left template A and right patch B ,

$$\text{ZNCC}(A, B) = \frac{\sum(A_i - \bar{A})(B_i - \bar{B})}{\sqrt{\sum(A_i - \bar{A})^2} \sqrt{\sum(B_i - \bar{B})^2}} \in [-1, 1], \quad (25)$$

invariant to affine brightness/contrast changes. It is evaluated at every 1-px sample of the epipolar curve (31×31 patches, vectorised); the peak is refined with the parabola through the scores (s_{-1}, s_0, s_{+1}) :

$$\Delta = \frac{s_{-1} - s_{+1}}{2(s_{-1} - 2s_0 + s_{+1})}. \quad (26)$$

9.3 Left–right consistency and Lucas–Kanade refinement

A match is accepted only if searching back from it along the *left* epipolar curve (obtained by inverting Γ_L) returns within 2px of the original click, its ZNCC is ≥ 0.6 and the template is textured. Occluded or ambiguous points fail this test. The accepted match is refined in 2-D by inverse-compositional Lucas–Kanade (translation model, 21×21 patch).

9.4 CLAHE

Both images are equalised with contrast-limited adaptive histogram equalisation: per-tile histograms clipped at $2\times$ the mean bin, excess redistributed, and the tile mappings blended *bilinearly between tile centres* (without blending, tile borders create false texture).

10 Uncertainty of a measurement

Click noise σ_{px} is propagated to first order through the *complete* model,

$$\sigma_{Z, \text{click}}^2 = \sum_{p \in \{u_L, v_L, u_R, v_R\}} \left(\frac{\partial Z}{\partial p} \right)^2 \sigma_{\text{px}}^2, \quad (27)$$

with central differences for $\partial Z / \partial p$. σ_{px} is estimated robustly from the epipolar residuals of the calibration ($1.4826 \cdot \text{MAD} / \sqrt{2}$ per click). The model’s own error ε (RMS relative LOOCV error) is added in quadrature: $\sigma_Z = \sqrt{\sigma_{Z, \text{click}}^2 + (\varepsilon Z)^2}$. The app displays $Z \pm \sigma_Z$.

11 Polynomial post-correction

A quadratic $Z_{\text{true}} \approx a + bZ_{\text{pred}} + cZ_{\text{pred}}^2$ fitted by least squares on a Vandermonde matrix is implemented (`fit_postcorrection`). Evaluated honestly (re-fitted inside every LOOCV fold) it *increases* the error ($5.1\% \rightarrow 6.9\%$): fitted on in-sample residuals of an already flexible model it learns noise. It is therefore disabled by default.

12 Evaluation

12.1 Protocol

With $N = 27$ labelled points a train/test split would waste data, so all real-data numbers are *leave-one-out* cross-validation (LOOCV): N fits on $N - 1$ points, each predicting the held-out point. Confidence intervals are from 4000 bootstrap resamples of the per-point errors. In-sample error is reported only for context; it is optimistic.

12.2 Real dataset

Model	LOOCV MAPE	95 % CI	worst point
v1 original (reproduced exactly)	10.08 %	5.7–16.3	77 %
Pinhole + disparity offset (2 params)	5.90 %	4.3–7.7	16 %
Pinhole + rotation (Γ frozen)	5.40 %	3.8–7.2	18 %
Γ grid, no rotation	8.22 %	5.4–11.2	28 %
Γ grid + rotation (v2)	5.14 %	3.3–7.2	21 %
v2 + quadratic post-correction	6.89 %	4.4–10.1	37 %

The paired bootstrap difference v1–v2 is 4.9 percentage points (95 % CI 0.7–10.4, $p \approx 0.008$). Three points (two near the right border of the image, one at 2.5m on the left) are $\approx 15\%$ off for *every* model, including the 2-parameter baseline, and two neighbouring points have opposite errors: this is label/click noise, not a model deficiency. The dataset therefore has a noise floor near 5 % that no model can beat; improvements below it cannot be resolved with 27 points. We also tested whether labels are Euclidean range $\|P\|$ rather than depth Z : depth fits better (5.1 % vs 5.9 %), so the labels are depths.

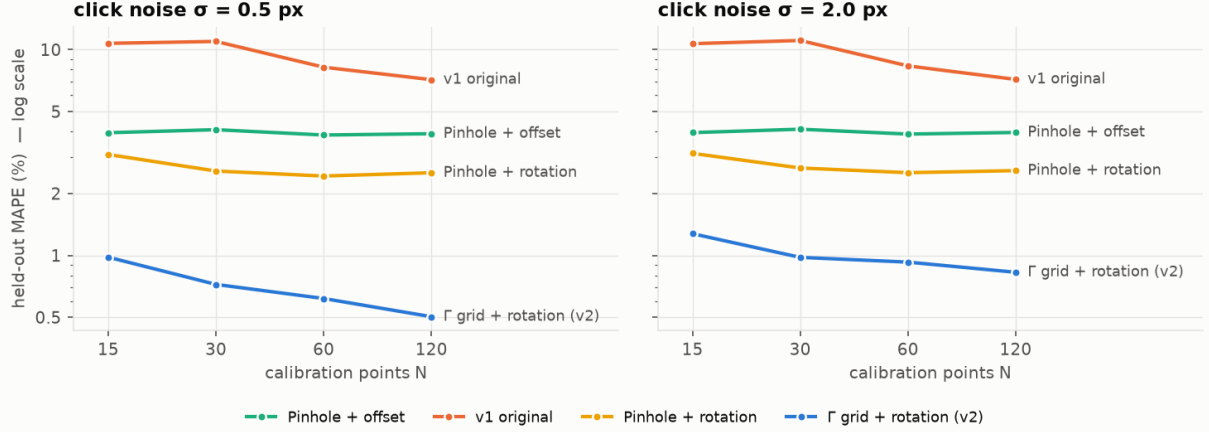
12.3 Controlled synthetic benchmark

To measure what the Γ grid is worth independently of label noise, we simulate a rig from a *different* model family: Brown–Conrady radial distortion ($k_1 = -0.10, k_2 = 0.04$ left; $-0.08, 0.03$ right), unequal focal lengths (2750/2790 px), off-centre principal points, baseline 0.20 m and rotation ($0.15^\circ, 0.9^\circ, -1.1^\circ$). Training points carry click noise σ and 1 % label noise; 500 held-out points are scored. Held-out MAPE (mean of 5 seeds):

$\sigma = 0.5$ px, $N =$	15	30	60	120
Pinhole + offset	3.95	4.09	3.85	3.90
v1 original	10.73	10.97	8.22	7.15
Pinhole + rotation	3.09	2.57	2.43	2.52
Γ grid + rotation	0.98	0.72	0.62	0.50

The Γ grid is the right idea: once the rotation is modelled it approximates real lens distortion 4–5 $\times$ better than a pinhole model and keeps improving with more data.

Synthetic rig with Brown-Conrady distortion (500 test points, mean of seeds)



12.4 Automatic matching

On a ray-traced scene (textured planes at 2, 3.5 and 6 m plus a floor) seen through the simulated rig, 200 random clicks: 80 % are accepted by ZNCC + left-right check; 92.5 % of accepted matches are within 3 % of the true depth (median error 0.6 %); 17 ms per click on a CPU.

13 Computational complexity

- Ray / interpolation: $O(1)$ per pixel. Triangulation: $O(1)$ (2×2 closed form).
- One training epoch: $O(N + |\Gamma|)$, fully vectorised; 600 epochs on 27 points take ≈ 0.3 s.
- LOOCV: N fits, ≈ 6 s for $N = 27$.
- Matching: $O(S P^2)$ for S curve samples and $P \times P$ patches, done twice (left-right); ≈ 17 ms per click at 2592×1944 .

14 Numerical stability

1. Parallel rays: $|\det A^T A| < 10^{-14}$ falls back to $Z = d_x/D$; $D \leq 0$ or $Z \leq 0$ returns NaN.
2. $\Gamma \in [0.05, 5]$ and $d_x \geq 0.02$ m prevent negative or infinite focal lengths and division by zero.
3. The relative loss avoids Z^2 gradient growth; Huber caps outlier gradients.
4. Gauge fixing removes the scale null-space; the anchor prior regularises unobserved cells.
5. Grid inversion clips intermediate pixels to the image $\pm 25\%$ and $\gamma \geq 10^{-6}$.

15 Limitations and future work

- **Data** is the bottleneck: 27 points with $\sim 5\%$ label noise. A calibration target (chessboard or ArUco board) at several known distances would give hundreds of sub-pixel points; the coverage overlay in the app (key V) shows which Γ cells lack data.
- **Bundle adjustment**: jointly refining 3-D point positions would allow using correspondences without known distance.
- **Multi-scale grids** ($3 \times 3 \rightarrow 9 \times 9$) could capture finer distortion once more data exists.
- Matching is local (single point); dense disparity (e.g. semi-global matching) is out of scope.

A Mapping between code and mathematics

Function (package <code>stereo_gamma</code>)	Implements
<code>geometry.bilinear_weights, interpolate_gamma</code>	eq. (3)
<code>geometry.rays</code>	eq. (2)
<code>geometry.rotation_with_jacobians</code>	$R(\omega)$ and $\partial R/\partial \omega_j$
<code>geometry.right_rays_in_left</code>	eq. (6)
<code>geometry.membrane_energy_grad</code>	E_{mem} and $\mathbf{L}\Gamma$
<code>model.StereoModel.triangulate</code>	OLS, eq. (7)
<code>model.StereoModel.triangulate_gd</code>	batch GD (+ Nesterov)
<code>model.StereoModel.project_left_point_to_right</code>	forward projection + grid inversion
<code>model.StereoModel.depth_uncertainty</code>	error propagation
<code>training.objective</code>	eq. (13) and all gradients
<code>training.huber, training.Adam</code>	Huber, Adam
<code>training.linear_init</code>	closed-form warm start
<code>training.fit_postcorrection</code>	quadratic post-correction
<code>matching.zncc_scores, parabolic_peak</code>	ZNCC, sub-pixel peak
<code>matching.epipolar_match</code>	epipolar search + left-right check
<code>matching.lk_refine, matching.clahe</code>	Lucas-Kanade, CLAHE
<code>evaluation.loocv, evaluation.metrics</code>	evaluation protocol
<code>legacy.train_legacy</code>	exact re-implementation of v1

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